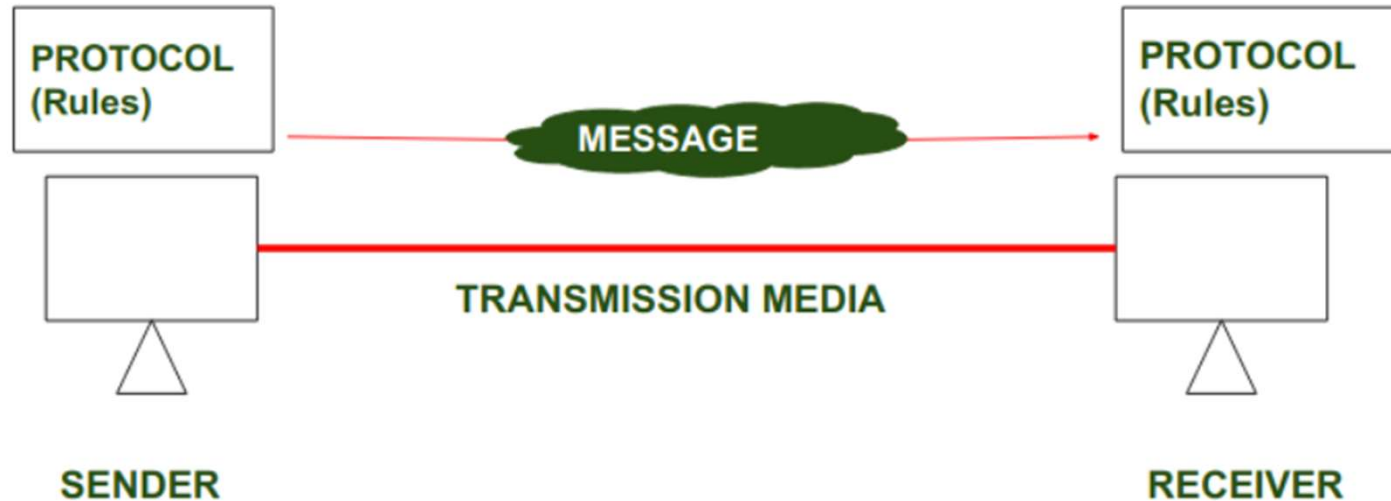


Protocols and Standards

What is Protocol?

- Protocol is a set of rules that governs data communications.
- It represents an agreement between the communicating devices.
- Different types of protocols are used for different types of communication.



Protocols and Standards

Key elements of protocol

1. *Syntax* - refers to the structure or the format of the data that gets exchanged between the devices.
2. *Semantics* – refers to the meaning of each section of bits
3. *Timing* – refers to two characteristics
 1. When data to be sent
 2. How fast it can be sent

Protocols and Standards

What is Standards?

- Standards are set of rules for data communication that are needed for the exchange of information among devices.
- It is important to follow standards which are created by various standard organizations like ISO, IEEE, ANSI etc.,

Types of standards

- **De Facto Standard:** The meaning of “*De Facto*” is “*By Fact*”.

Standards not been approved by any organization but adopted

- **De Jure Standard:** The meaning of “*De Jure*” is “*By Law*” .

Standards have been approved by officially recognized bodies

Network Models

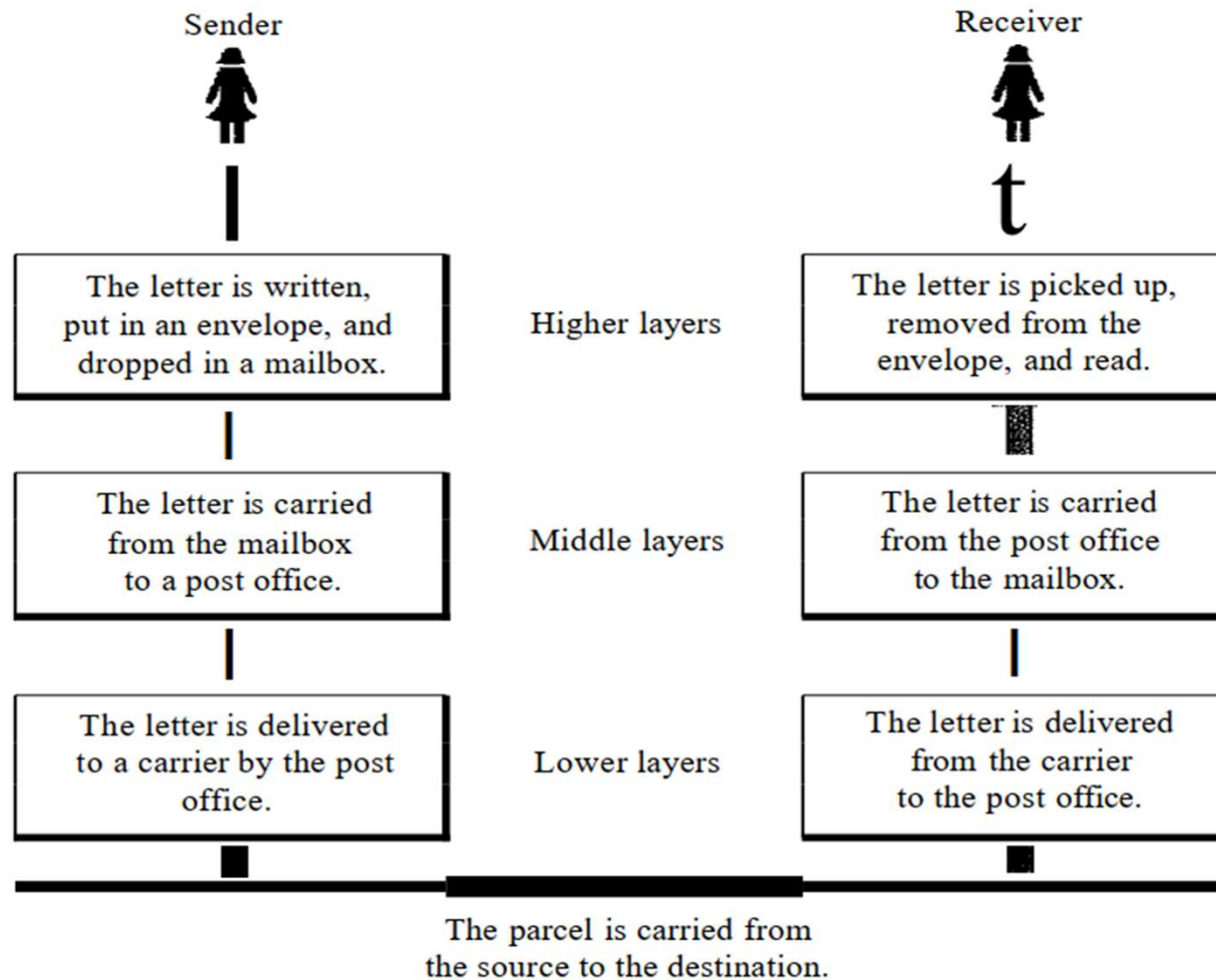
A network is a combination of hardware and software that sends data from one location to another

Hardware : The hardware consists of the physical equipment that carries signals from one point of the network to another.

Software : The software consists of instruction sets that make possible the services that we expect from a network.

In computer networks, the network models shows different activities of networks through layered architecture.

Layered Tasks



Protocol Layering

1. Client – Server Model
2. Transmission Control Protocol/Internet Protocol(TCP/IP) suite
3. Open System Interconnection(OSI) Model

OSI MODEL

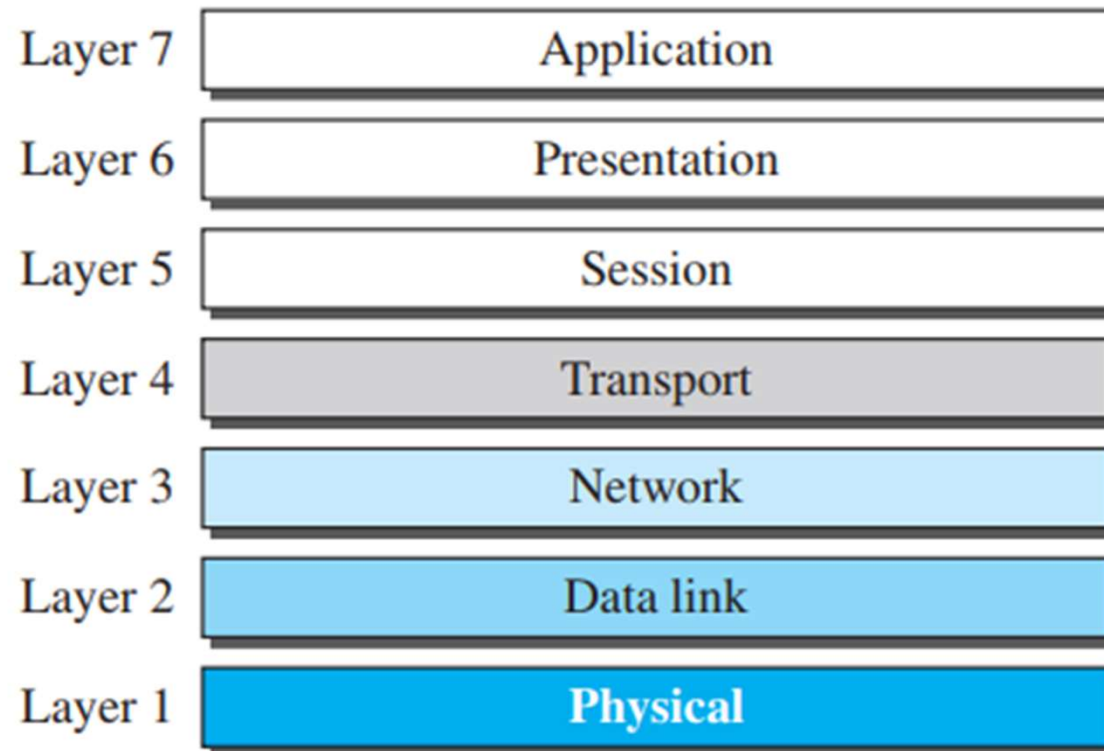
What is ISO ?

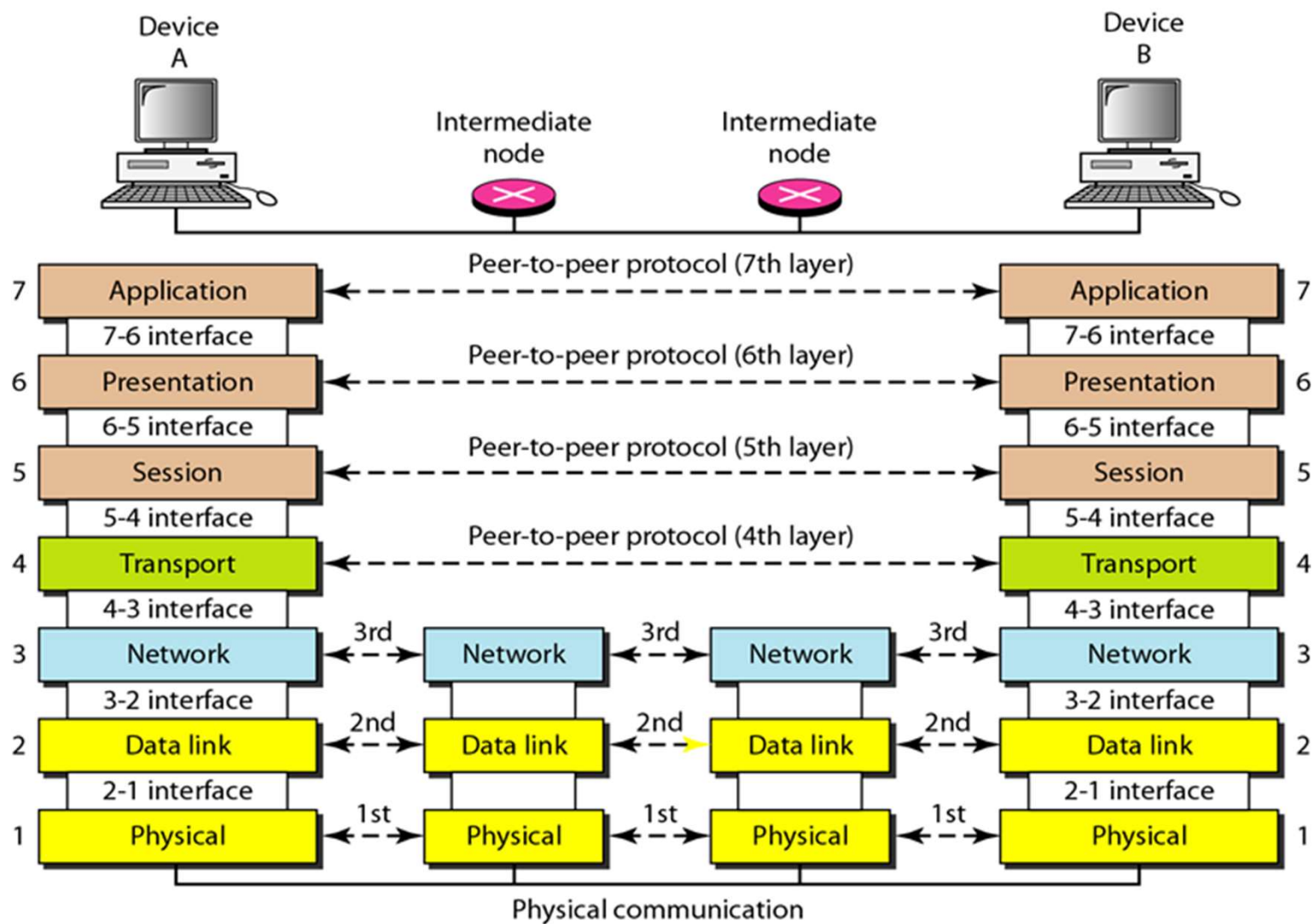
Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.

What is OSI ?

An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late **1970s**.

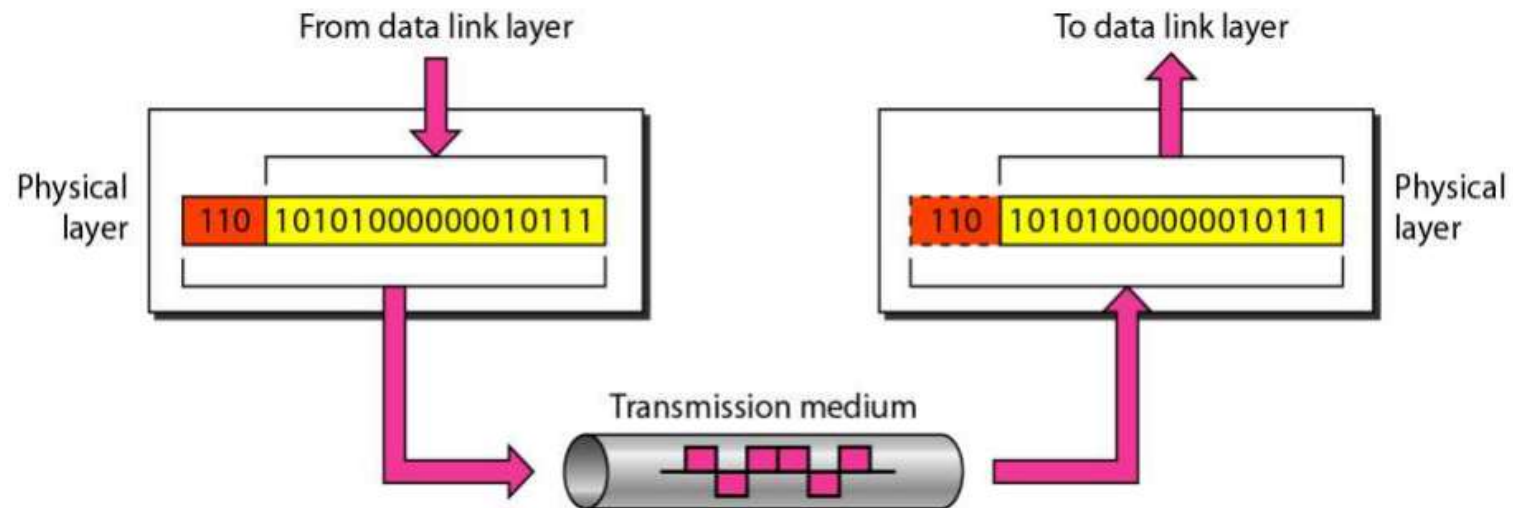
Layered Architecture





PHYSICAL LAYER

- ❖ This layer includes the **physical equipment** involved in the data transfer, such as the cables and switches.
- ❖ This is also the layer where the data gets converted into a **bit stream**, which is a string of 1s and 0s.
- ❖ The physical layer of both devices must also agree on a **signal convention** so that the 1s can be distinguished from the 0s on both devices.



The physical layer is responsible for movements of individual bits from one hop (node) to the next

Physical layer

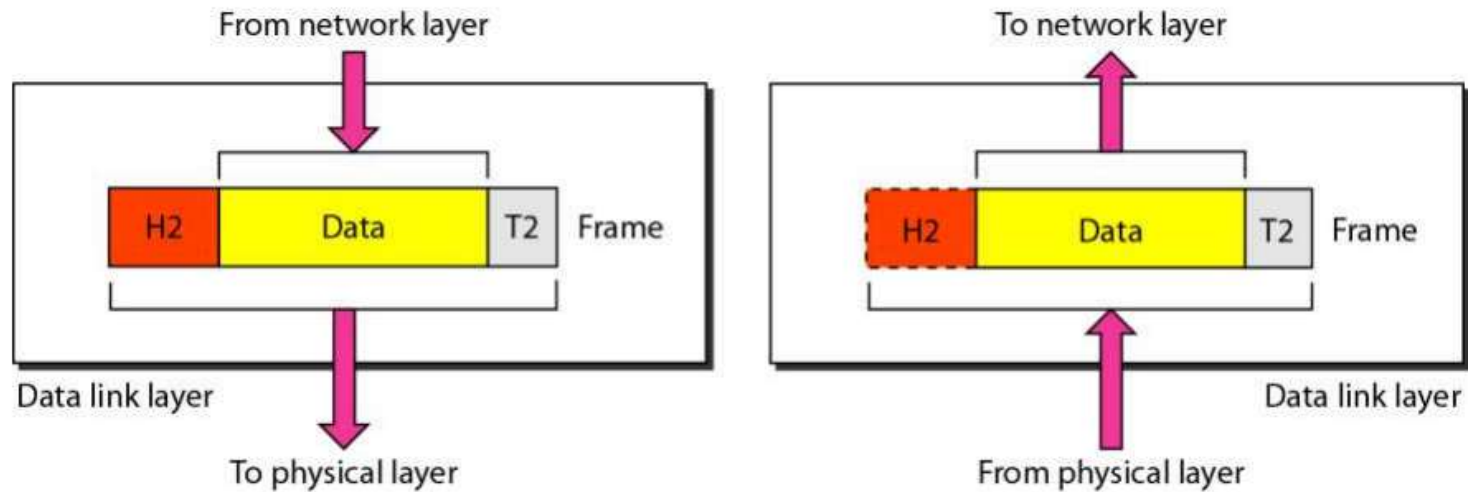
Responsibility of Physical layer:

1. Physical characteristic of interface and medium
2. Representation of bits
3. Data/Transmission rate
4. Synchronization of bits
5. Line configuration
6. Physical topology
7. Transmission mode

Data link layer

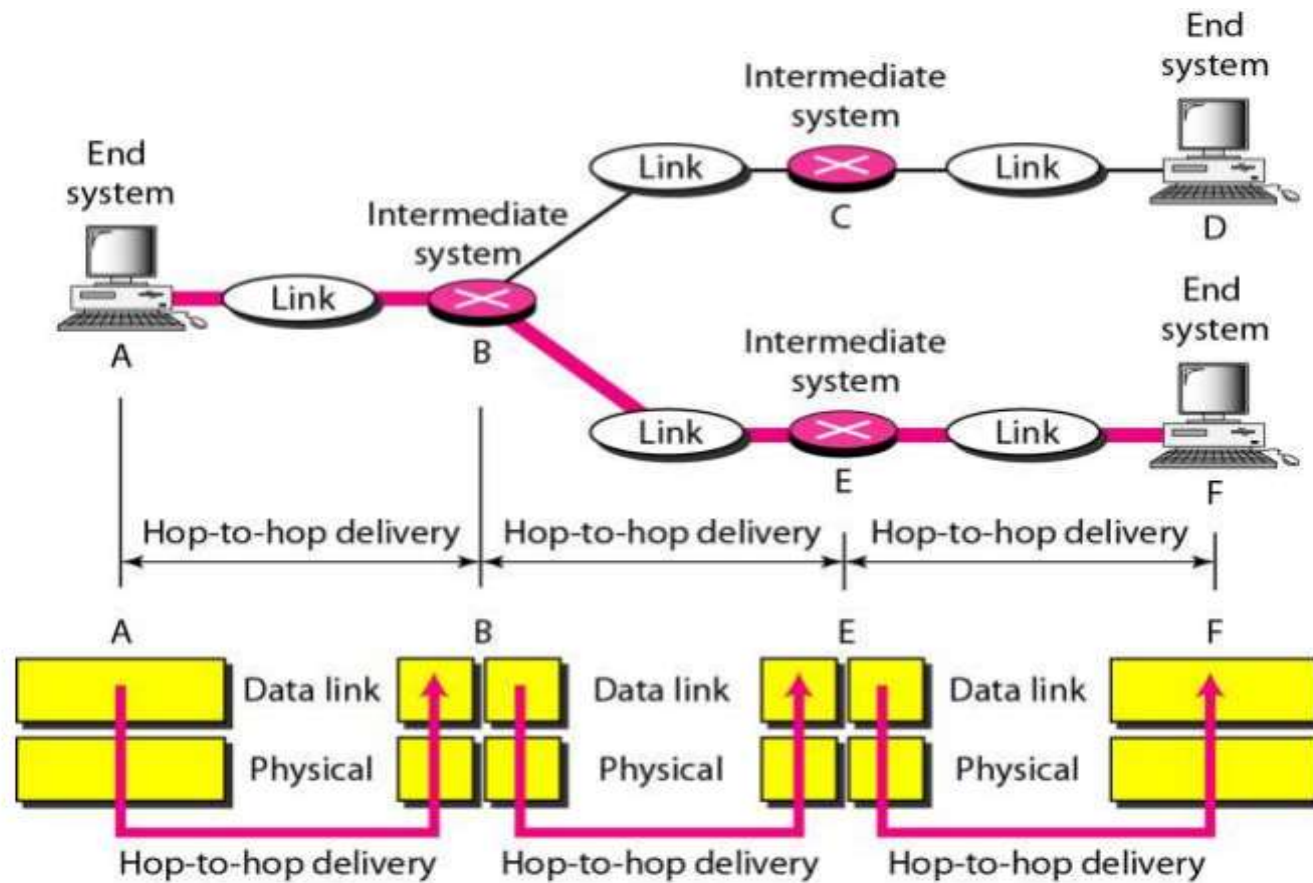
- ❖ The data link layer transforms the physical layer, a raw transmission facility, to a reliable link.
- ❖ The data link layer takes packets from the network layer and breaks them into smaller pieces called frames.
- ❖ The data link layer is responsible for flow control and error control in intra-network communication.
- ❖ It makes the physical layer as error-free to the upper layer.

Data link layer



The data link layer is responsible for moving frames from one hop (node) to the next

Hop-to-hop delivery



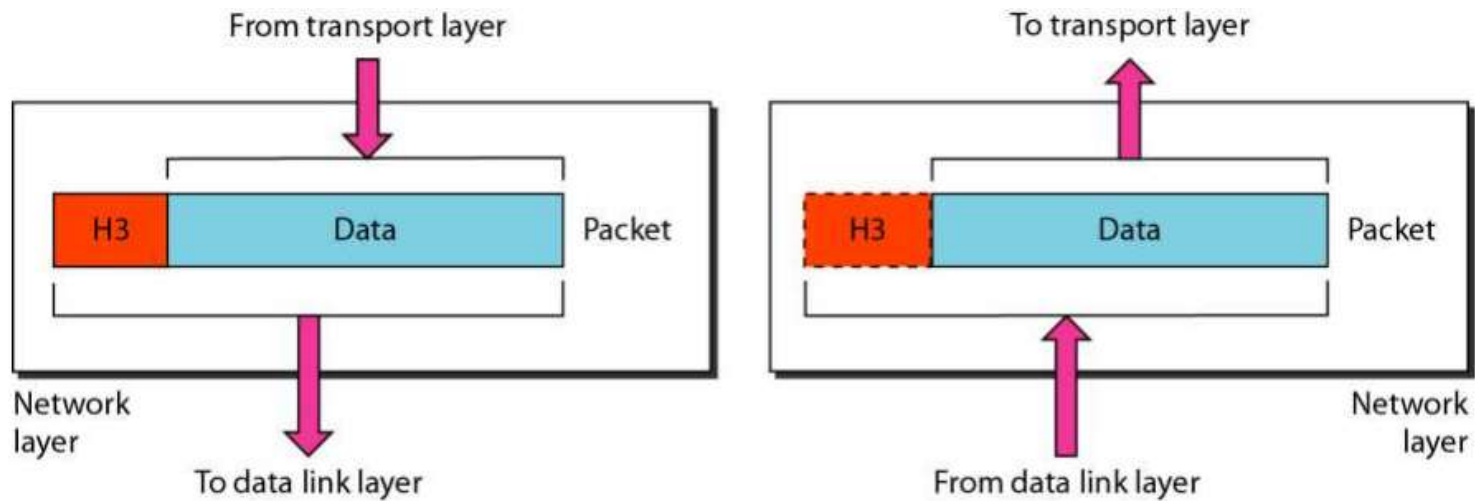
Data Link Layer

Responsibilities of Data Link Layer:

- ❖ Framing
- ❖ Physical addressing
- ❖ Flow control
- ❖ Error control
- ❖ Access control

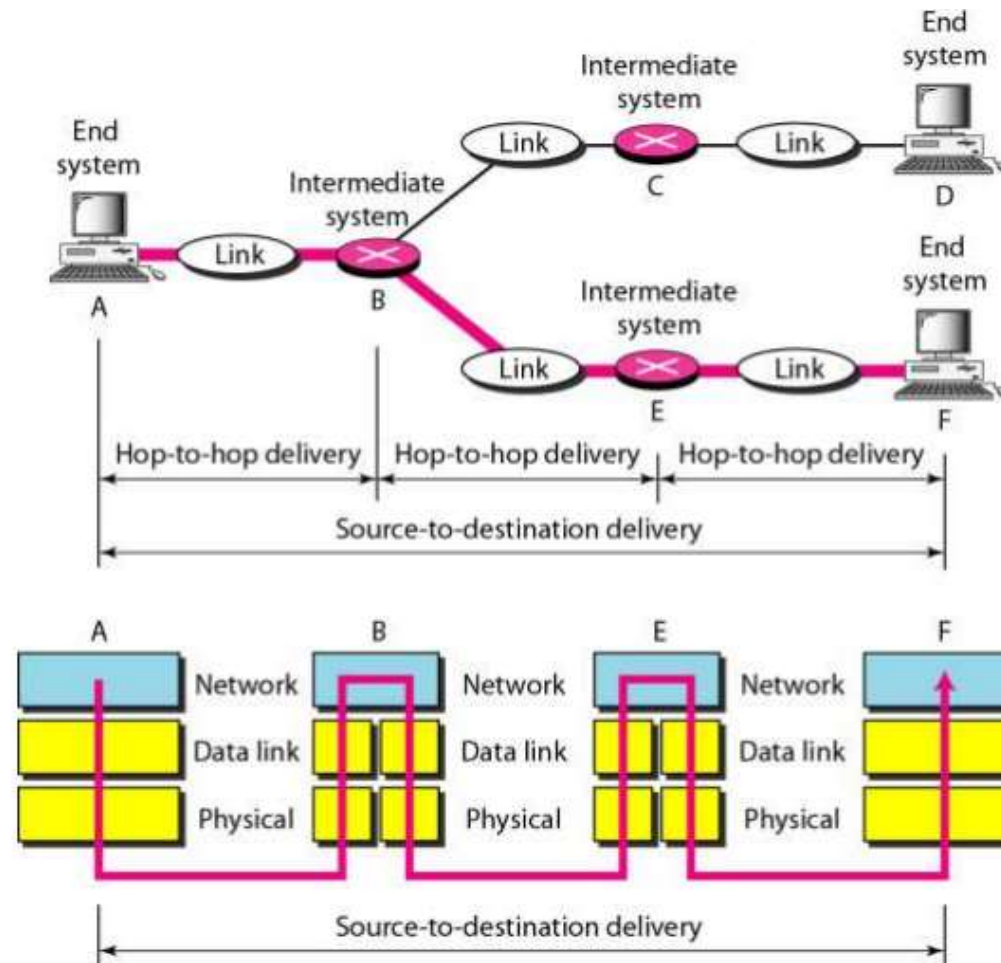
Network layer

- ❖ The network layer is responsible for facilitating **data transfer** between **two different networks**.
- ❖ If the two devices communicating are on the same network, then the network layer is unnecessary.
- ❖ The network layer breaks up **segments** from the transport layer into smaller units, called **packets**, on the sender's device, and reassembling these packets on the receiving device.
- ❖ The network layer also **finds the best physical path** for the data to reach its destination; this is known as **routing**.



*The network layer is responsible for the **delivery of individual packets** from the **source** host to the **destination** host*

Source-to-destination delivery



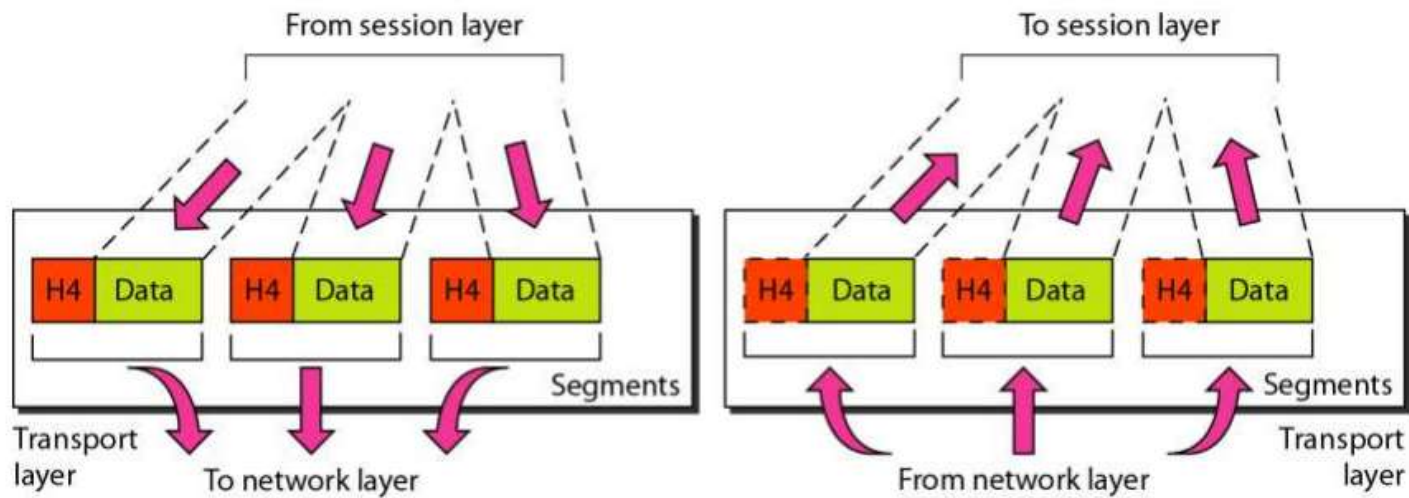
Network Layer

Responsibilities of Network Layer:

- ❖ Logical addressing
- ❖ Routing

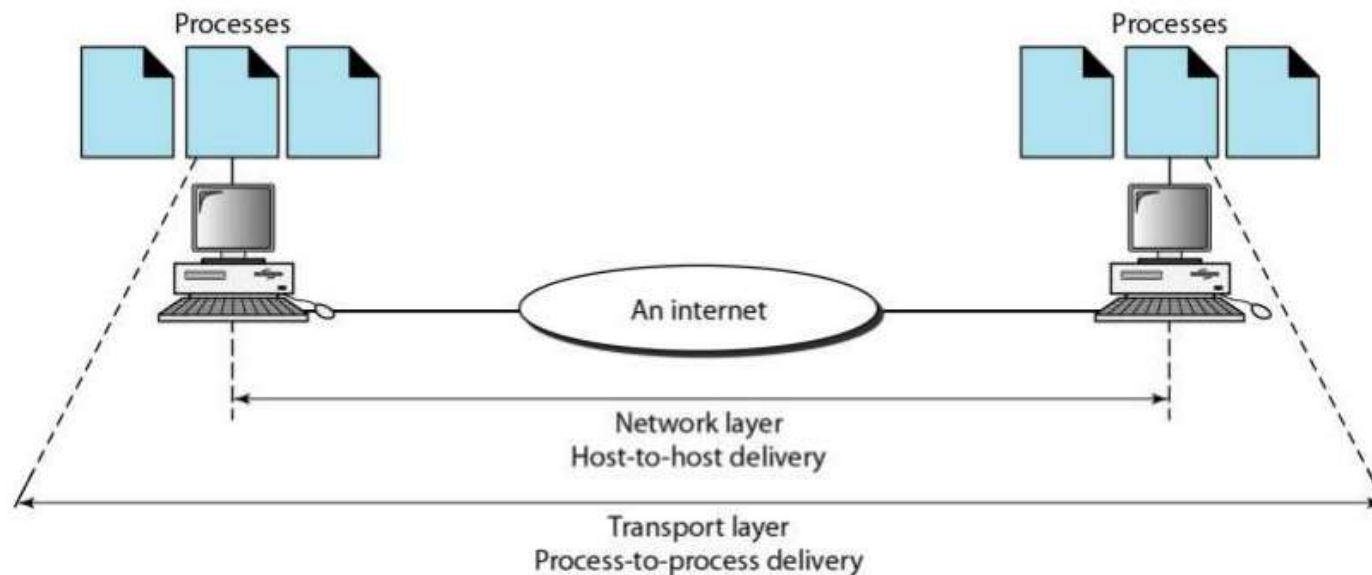
Transport layer

- ❖ Layer 4 is responsible for **end-to-end communication** between the two devices. This includes taking data from the session layer and breaking it up into chunks called **segments** before sending it to layer 3.
- ❖ The transport layer on the receiving device is responsible for reassembling the segments into data the session layer can consume.
- ❖ The transport layer is also responsible for **flow control and error control**.
- ❖ **Flow control** determines an **optimal speed of transmission** to ensure that a sender with a fast connection doesn't overwhelm a receiver with a slow connection.
- ❖ The transport layer performs **error control** on the receiving end by ensuring that the **data received is complete**, and requesting a retransmission if it isn't.



The transport layer is responsible for the delivery of a message from one process to another process.

Reliable process-to-process delivery of a message



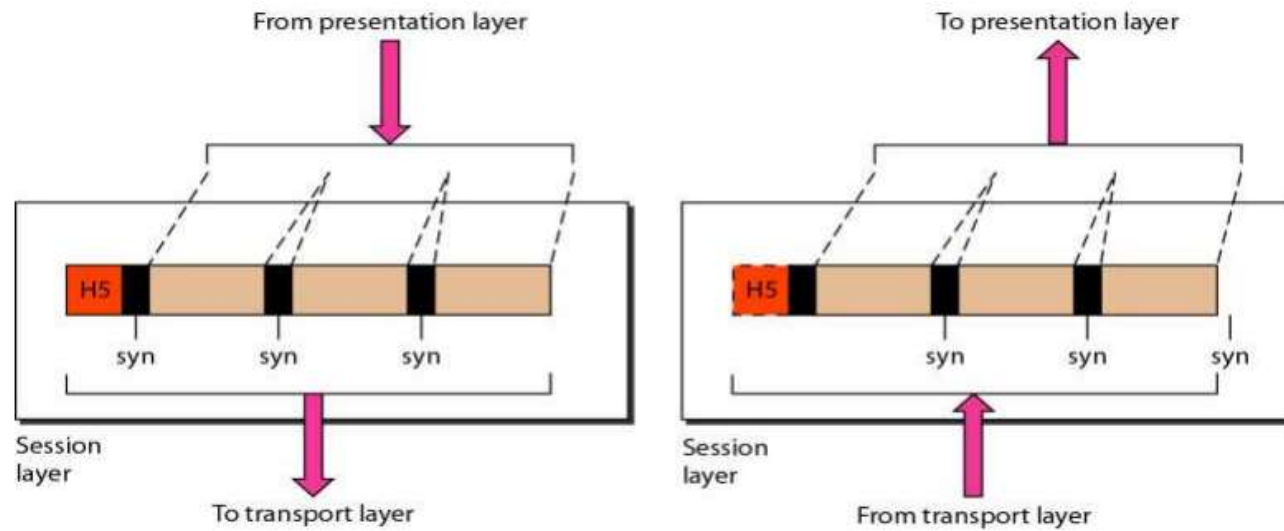
Transport Layer

Responsibilities of transport layer:

- ❖ Service point addressing
- ❖ Segmentation and reassembly
- ❖ Connection control

Session layer

- ❖ This is the layer responsible for **opening and closing communication** between the two devices.
- ❖ The time between **when the communication is opened and closed** is known as the session.
- ❖ The session layer ensures that the session stays open long enough to transfer all the data being exchanged, and then promptly closes the session in order to avoid wasting resources.
- ❖ The session layer also synchronizes data transfer with checkpoints.

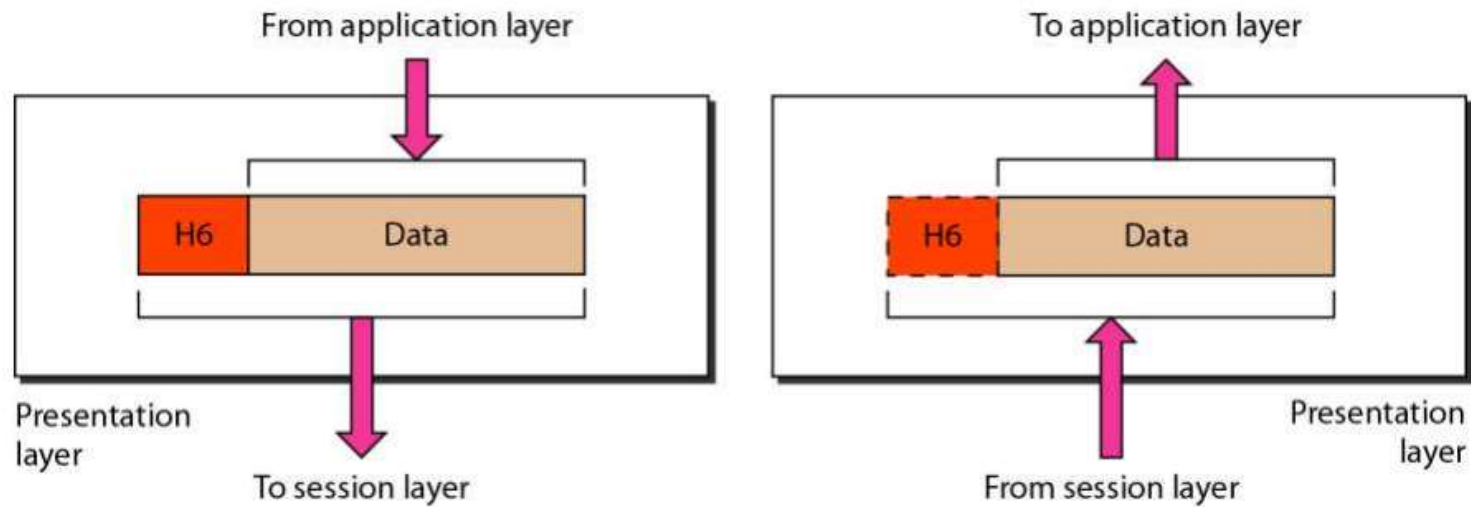


The session layer is responsible for dialog control and synchronization

Presentation layer

- ❖ The presentation layer is responsible for **translation, encryption, and compression** of data.
- ❖ Two devices communicating may be using different encoding methods, so layer 6 is responsible for **translating incoming data into a syntax (UTF-8)** that the application layer of the receiving device can understand.

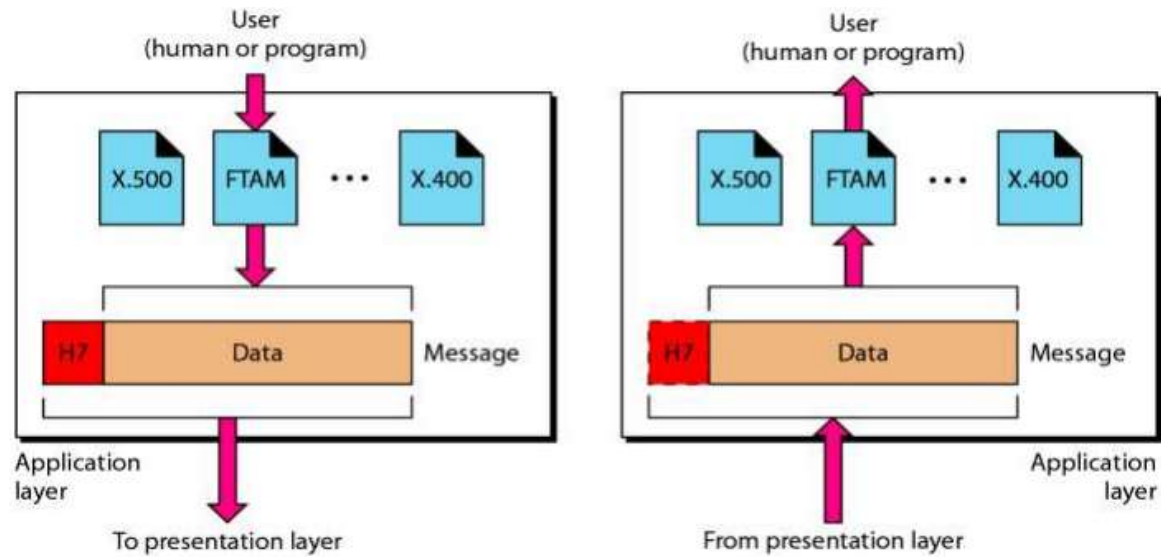
- ❖ If the devices are communicating over an encrypted connection, layer 6 is responsible for **adding the encryption (SSL or TLS) on the sender's end as well as decoding the encryption on the receiver's end** so that it can present the application layer with unencrypted, readable data.
- ❖ Finally the presentation layer is also **responsible for compressing data (Lossy and Lossless)** it receives from the application layer before delivering it to layer 5.
- ❖ This helps improve the speed and efficiency of communication by minimizing the amount of data that will be transferred.



The presentation layer is responsible for translation, compression, and encryption

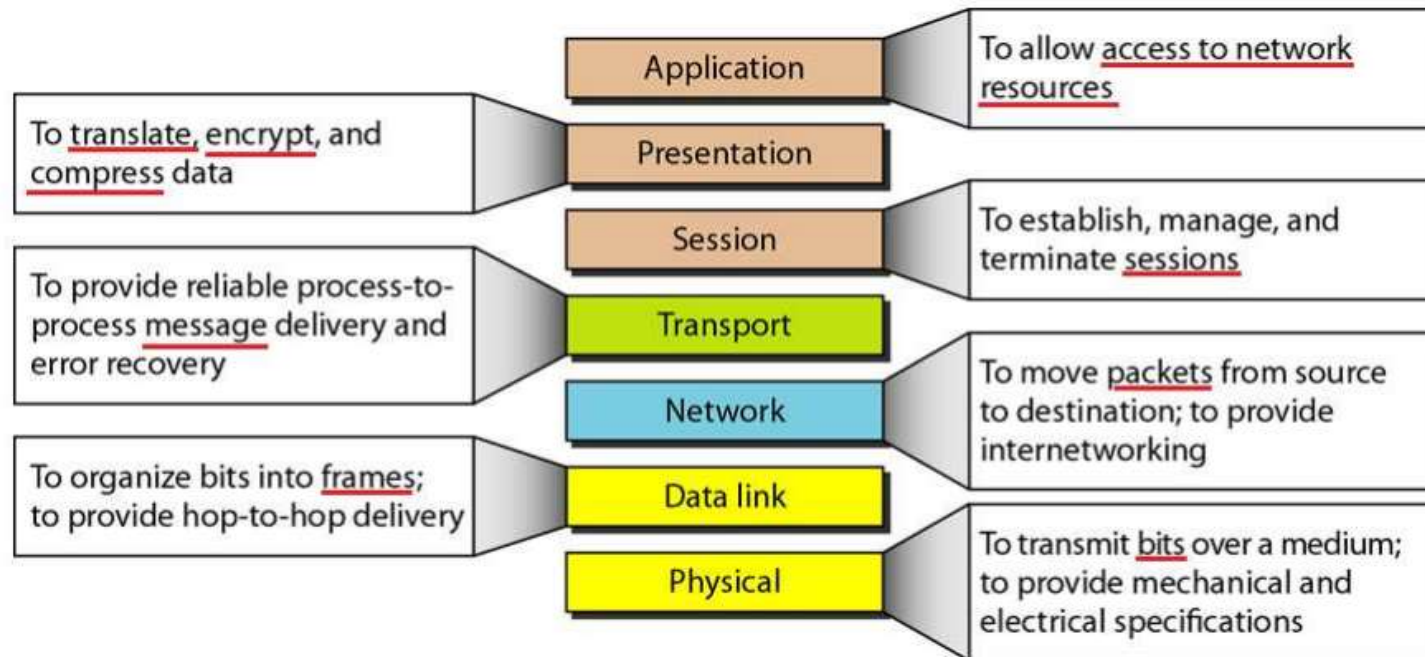
Application layer

- ❖ This is the only layer that **directly interacts with data from the user**. Software applications like web browsers and email clients rely on the application layer to initiate communications.
- ❖ But it should be made clear that client software applications are not part of the application layer; rather the application layer is responsible for the protocols and data manipulation that the software relies on to present meaningful data to the user.
- ❖ Application layer protocols include HTTP as well as SMTP (Simple Mail Transfer Protocol is one of the protocols that enables email communications).



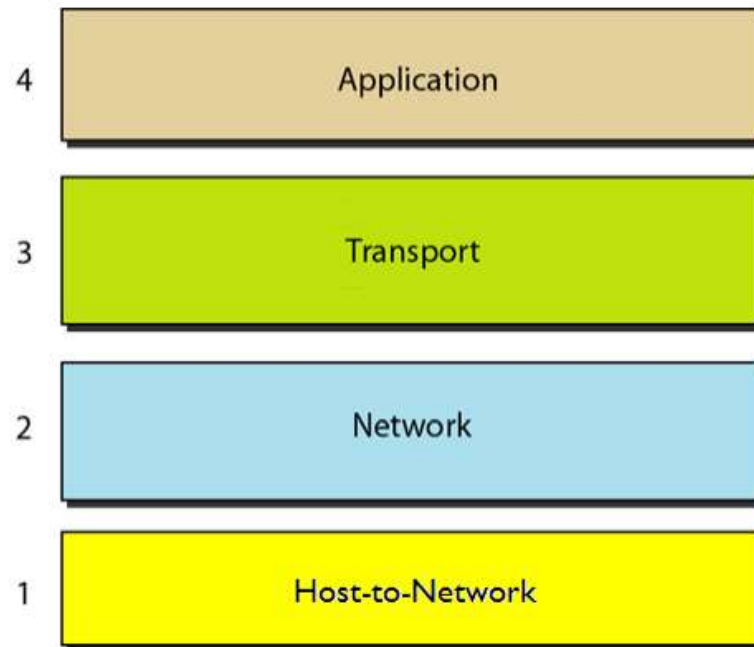
The application layer is responsible for providing services to the user

Summary of layers

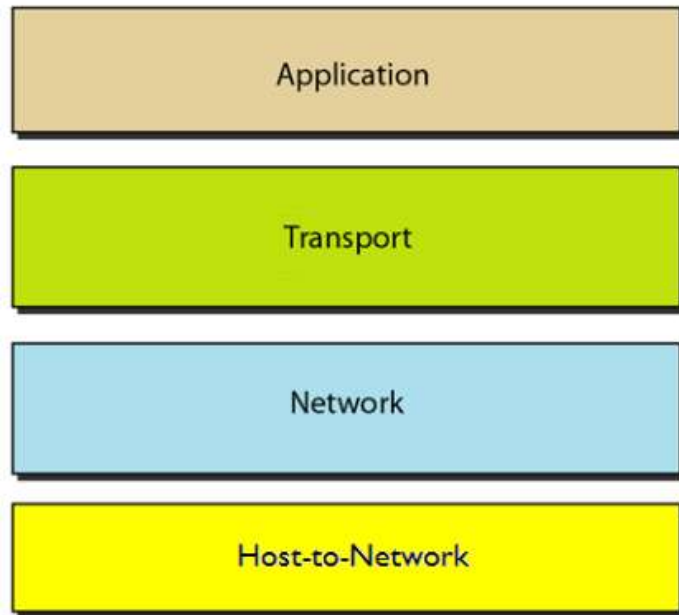


TCP/IP PROTOCOL SUITE

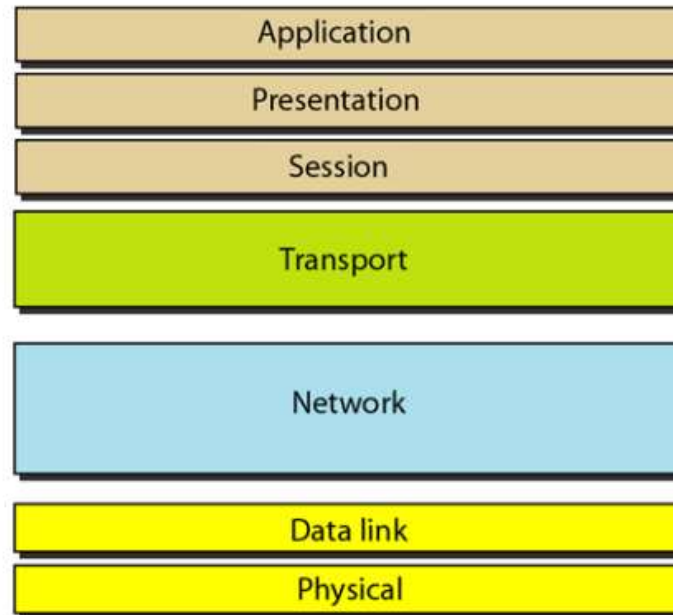
The original TCP/IP protocol suite was defined as having four layers : host-to-network, network(internet), transport and application



- ❖ The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- ❖ However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: *physical, data link, network, transport, and application*



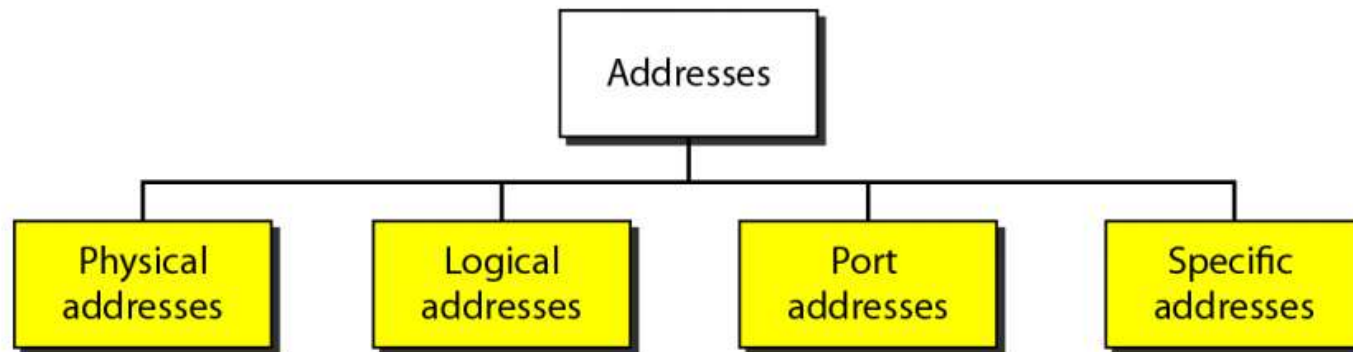
TCP / IP Model



OSI Model

ADDRESSING

Four levels of addresses are used in an internet employing the TCP/IP protocols:
physical, logical, port, and specific



Relationship of layers and addresses in TCP/IP (Internet Model)

